An Advisory Report

From Capture to Care: Initiating Responsible Web Archiving at NIOD

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Summary

This case study explores the possibilities, challenges, and institutional implications of initiating web archival practices at the Netherlands Institute for War, Holocaust and Genocide Studies (NIOD). This project emerged from an organisational goal to expand its collections with records on contemporary armed conflicts, mass violence, and genocide. While NIOD has a longstanding tradition of collecting analogue sources related to the Second World War, current day events are increasingly documented online. This report explores possibilities for transformation from predominantly donation-based, analogue collecting toward a proactive, digital strategy grounded in web archiving.

Drawing on a literature review, stakeholder and tension mapping, and interviews, the study conceptualizes web archiving not as a technical solution aimed at comprehensive capture, but as an ongoing archival process shaped by selection, governance, ethical responsibility, and transparency. Archival scholarship informs us that digital abundance intensifies rather than resolves the need for appraisal. Legal constraints and platform infrastructures ensure that incompleteness is an inherent condition of web archives. Ethical frameworks, particularly feminist ethics of care, foreground the responsibilities archives have toward subjects in records and affected communities whose traces might be preserved.

The research shows how web archiving decisions are structured by recurring tensions, including those between abundance and selection, visibility and vulnerability, speed and care, and privacy and transparency. These tensions create a need for institutional choices concerning scope definition, collection methods, access regimes, documentation practices, and sustainability. To explore how such decisions interact over time, the project introduces "Making the Archive", a web archiving decision simulator developed in Twine. The simulator functions as a reflective research tool that models trade-offs between the archival values of ethics, sustainability, legality, transparency, and completeness. Rather than prescribing an optimal solution, it shows that arguments can be made for multiple decisions in the process of creating a web archive.

Our recommendations are therefore not structured as a single, fixed web archiving strategy that aims for comprehensive capture. Instead, we suggest approaching web arching as a gradual, reflective institutional practice that develops through documented decisions, experimentation, and ongoing evaluation. At the end of the report, we provide procedural starting points that outline how NIOD could begin engaging with web archiving in an accountable and manageable way.

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Introduction

The NIOD Institute for War, Holocaust and Genocide Studies has a long tradition of research and collection-building on the topics of world wars, mass violence, and genocide, including their long-term societal consequences. Central to both its research agenda and its archival collection is the Second World War. Since its establishment as the National Bureau for War Documentation in May 1945, the NIOD's holdings on this period have grown through active selection in the early postwar years and have continued to expand through donations up to the present day ("The History of NIOD," n.d.).

Even though war and genocide studies encompass a wide arrange of topics, such as contemporary armed conflicts, ongoing instances of mass violence, and modern manifestations of genocide, these remain underrepresented in NIOD's collection. These conflicts are increasingly documented in digital environments, where information is produced, circulated, and contested at unprecedented speeds. Documenting current day conflicts therefore requires moving beyond relying primarily on analogue sources and passive acquisition strategies.

One of the strategies put forward by the NIOD to tackle this problem is web archiving. By systematically capturing online sources web archiving enables institutions to document contemporary wars and genocides in real time and to preserve a wide range of perspectives on selected topics. In this context the digital collections department of NIOD has asked us to give advice on and provide guidelines for their first steps towards web archiving of current day conflicts. This report will analyse what is necessary for a transition from a primarily analogue, donation-orientated tradition of collecting towards a more proactive collection strategy of web-based materials.

The recommendations in this report will be based on an extensive literature review on the topic of web archiving, which provides theoretical context for a case study of the NIOD. We analyse the institution through stakeholder analysis, identifying key parties involved and tensions between them. Additionally, we conducted interviews with NIOD staff to get a better understanding of its organisational structure, aims and goals, as well as the institute's practical possibilities and limitations. Central themes throughout the analysis include selection scope, collection strategies, and the importance and practicality of transparency.

Finally, we developed a web archiving decision simulator titled, *Making the Archive*. Designed to communicate our research in an interactive format, the simulator models key dilemmas involved in setting up a web archive and highlights trade-offs embedded in different

strategic choices. In this report we discuss how the simulator functions, as well as its intentions and limitations. This ultimately leads to our outcomes, recommendations, and conclusion.

Methodology

We adopted a qualitative, multi-method research design to examine how web archiving reshapes archival responsibility in the context of the NIOD. This approach combines conceptual analysis, empirical inquiry through interviews, and scenario-based exploration in order to foreground decision-making and trade-offs in initiating web archiving practices.

1. Initially we conducted a **literature review** focused on archival theory and web archiving scholarship. This review **informed the conceptual framing** of the project, particularly around archival selection, digital abundance and ephemerality, ethics of care, transparency, and governance. The literature was used analytically to identify recurring challenges and concepts, which subsequently structured the empirical and design-based components of the research.
2. Our group performed a **stakeholder and tension mapping** exercise which did not aim to produce a fixed stakeholder model, but to identify relationships, hierarchies, and problem areas shaping web archiving at NIOD. The mapping helped clarify how institutional mandates, legal frameworks, technical infrastructures, creators, and users intersect; and where conflicts between values such as completeness, sustainability, legality, and care arise.
3. With the approach of **semi-structured** style, we organized **interviews** with staff members from NIOD's collections and digital documentation department. We focused the interview questions on current practices, institutional constraints, emerging discussions around web-based material, and considerations related to selection, transparency, ethics, and sustainability. These interviews provided insight into how theoretical concerns identified in the literature manifest in institutional practice and helped ground our project in NIOD's specific context.
4. Our project adopts a scenario-based, design-led research component in the form of an **interactive decision simulator.** The simulator translates insights from the literature, stakeholder mapping, and interviews into a sequence of institutional choices related to scope, collection methods, access, and documentation. As opposed to functioning as a predictive or solution-oriented tool, the simulator operates as an analytical device to

explore how archival values are negotiated over time and how consequences come up through cumulative decisions.

Together, these methods allow the project to move between theory, institutional practice, and exploratory design. Through the combination of these methods, we seek to make visible how responsibility, constraint, and judgement are produced through institutional processes. This methodological approach reinforces web archiving as an ongoing practice shaped by cumulative decisions rather than isolated choices, thereby providing the foundation for our analysis of NIOD's current position and the implication of initiating web archiving practices.

Conceptual Framing; Web Archiving as Archival Process and Responsibility

In this project, we frame web archiving not as a technical task aimed at comprehensive capture, but as an institutional process shaped by selection, responsibility, and governance. Drawing on archival theory and web archiving scholarship, we position web archiving as a site where institutional values are negotiated through decisions, rather than resolved through singular solutions.

Archival theory has long challenged the notion of archives as neutral repositories that merely preserve what survives. Within the post-custodial archival paradigm, scholars such as Terry Cook (1997) argue that appraisal and selection are not marginal activities but constitute the core of archival practice, actively moulding historical knowledge and collective memory (p.22). From this perspective, archives do not simply reflect the past but participate in its construction by defining what counts as evidence. Such responsibility is not diminished in digital environments. Currently, NIOD is experiencing a shift from a donation-driven institution to one that needs to make a proactive effort to collect web archiving material; the relevance of this reflects on how the shift makes appraisal more visible and contestable. As Cook notes, "with electronic records, the physical medium becomes almost totally irrelevant, as the records themselves will be migrated forward long before the physical storage medium deteriorates" (p.45). Digital records therefore shift archival concern away from material decay toward long-term stewardship, sustained decision-making, and explicit accountability for selection.

Web archiving scholarship further demonstrates how digital environments intensify these responsibilities. Ian Milligan (2019) emphasizes that histories and memory institutions can no longer ignore "the phenomenal transformation of information and records of experience online;" underscoring the necessity of engaging with web archives as historical sources (p.14). A defining feature of these environments is abundance; "such digital traces are now so ubiquitous that finding something specific among the available information is the real challenge" (p.9). Thus, the context of being in need of materials no longer structures archival frustration; instead, super-abundance makes selectivity unavoidable. At the same time, the web is characterized by ephemerality. As Milligan cautions, "web archives are not only necessarily incomplete," and that such absences are not inherently curatorial failures but are often shaped by ethical, legal, and technical constraints (p.16). Building on this framing, we situate incompleteness as an inherent condition of web archiving rather than a problem to be solved. Thereby, the pressure on NIOD to define what to select intensifies, moving the institution away from solely focusing on content and toward accepting incompleteness as an archival condition. In this sense, NIOD must engage with the value of incompleteness as part of its design and documentation strategies.

Extending this understanding, Niels Brügger and Ralph Schroeder (2017) establish web archives as designed and mediated objects, as alternative to, transparent reflections of online activity. They argue that "the web is not becoming whole, but rather a series of clusters," influenced by linguistic factors, state policies, and platform-specific logics (p.4). Consequently, web archiving strategies – whether selective, thematic, or large-scale – actively produce particular representations of the past. NIOD has mainly practiced selective archiving, primarily focusing on World War II collections. Their branching out towards contemporary conflicts, particularly on the web, does not introduce selectivity as a principle, but shifts how collecting should be approached. Thus, collection strategy becomes a part of NIOD's institutional positioning: what 'NIOD frames as evidence' of conflicts. Additionally, a vast majority of the institution's collection can be accessed on-site in their reading room in Amsterdam. In Brügger and Schroeder's words, "a collection without access does not make much sense," yet access is uneven and often restricted by legal and institutional frameworks (p.10). Hence, this highlights how collection strategies, access regimes, and documentation practices are inseparable from questions of governance and institutional positioning.

Ethical considerations are central to this framing, particularly for institutions engaging with histories of war, genocide, and violence. NIOD primarily functions as a research institution, meaning that the main users in mind are researchers. Taking into consideration how

NIOD is reaching out to affected communities through interviews, and increasingly archiving present-day conflicts, this implies that access design and documentation must be considered as record subjects and potential future users. Michelle Caswell and Marika Cifor (2016) propose a shift from predominantly legalistic, rights-based approaches toward a feminist ethics of care, in which archivists are understood as care givers "bound to records creators, subjects, users, and communities through a web of mutual affective responsibility" (p.23). They argue that an archival approach grounded in radical empathy requires that affected communities become "central focal points in all aspects of archival endeavour, from appraisal to description to provision of access" (p.24). In digital contexts, where individuals often do not anticipate archival preservation, this arrangement foregrounds the ethical implications of visibility, exposure, and long-term use.

Finally, Schafer and Winters (2021) underline that web archives are inherently political infrastructures. They observe that web archives are "surprisingly poor at consistently documenting their own histories and practices," (p.138) making it difficult to assess their scope, limitations, and biases. Transparency, in this sense, moves beyond access to include the documentation of selection criteria, technical constraints, and institutional rationales. Similarly, they argue that web archives "do have politics," (p.134) shaped by the values and imaginaries embedded in their design and governance. This project sheds light on this call for memory institutions to move away from solely considering the content of archival material, towards acknowledging and engaging with the questioning and documentation of archival practices. As NIOD is deeply rooted in holding archival material which cannot be separated from politics, this report supports making guidelines, metadata, and decision logs public-facing whenever possible.

Together, these perspectives establish web archiving as ongoing institutional practice constantly formed by selection, mediation, ethical responsibility and transparency. This conceptual framing informs the analysis of NIOD's current position and underpins our emphasis on process rather than completeness. Instead of asking how NIOD can archive everything, this framework shifts attention to how web archiving decisions are made, justified, and documented as part of responsible archival practice.

Case Study: NIOD and Web Archiving

"The stakes are high. If we do not come to grasps with web archives, the histories we will write will be fundamentally flawed" (Milligan 2016, p.78). While Milligan's urgency to shed light

on what gets lost if institutions look past the web, we first look at the NIOD's gaps and current archival practices. As established in the introduction, the NIOD operates at the intersection of historical research, public memory, and large-scale violence. Although still being in the process of getting accustomed to the web, its mandate has gradually expanded beyond a primary focus on the Second World War toward a broader, comparative engagement with mass violence in the twentieth and twenty-first centuries. NIOD itself emphasizes recurring research themes including: "perpetratorship, memories, and trauma, transitional justice, restoration of rights, regime changes and behaviour and experiences;" reflecting an ambition to study mass violence in a multidisciplinary and international framework ("The History of NIOD," n.d.). This expansion in research scope has historically been accompanied by active collecting practices. From its early years, NIOD staff has not only received personal documents such as diaries, correspondence, and photo albums, but also actively search for relevant material themselves.

At the same time, NIOD is institutionally embedded within a broader governance framework. As a research institute under the Royal Netherlands Academy of Arts and Sciences (KNAW), and indirectly funded by the Ministry of Education, Culture and Science (OCW), NIOD operates within cultural policy frameworks that highlight accessibility, plurality, quality, and public accountability. Initiatives such as the *Netwerk Digitaal Erfgoed* (NDE), supported by OCW, further signal an expectation that heritage institutions should engage in digital collaboration and interoperability (Royal Netherlands Academy of Arts and Sciences 2024, p.67). This institutional positioning creates both opportunities and constraints for web archiving. While NIOD has strong research legitimacy and access to digital preservation infrastructure through KNAW, its role as a publicly funded memory institution also heightens legal, ethical, and governance responsibilities when engaging with contemporary, web-based material.

Stakeholders: Who is considered? Who is in control?

Web archiving at NIOD is shaped by a constellation of stakeholders whose interests, responsibilities, and expectations intersect but do not fully align. At the institutional level, OCW and KNAW outline the conditions under which web archiving can take place through funding structures, legal frameworks, and infrastructural support. Researchers form a second key stakeholder group, while their interest in what the NIOD preserves and makes accessible is rather high, their power to intervene in these processes remains relatively limited. With contemporary conflicts and debates being documented online, this group is simultaneously

reliant on web-based sources and on institutional selection decisions that determine what is preserved.

Stakeholder name	Power	Roles	Interest (in web-archiving)
Ministry of OCW	High	Funding of educational, cultural and research institutions	Medium
KNAW	High	Governance of subsidiary institutions	High
NIOD	Very High	Research, collecting, archiving, community-building, publishing	High
State media broadcasters	Medium	Publishing media on newsworthy subjects, making use of archival footage and evidence	Low-Medium
Independent media companies	Medium	Publishing media on newsworthy subjects; consideration of funding structures, making use of archival evidence	Low-Medium
Researchers	Medium-Low	Historic & cultural research, consulting archives	High
Independent journalists	Medium-Low	Investigative journalism, using archival evidence	Medium-High
Affected communities	Low	Implicated and represented in collection, aiding in contextualization, interest in archival evidence	High
General Public	Low	Audience for publicly accessible parts of collection, stakeholders in institutional accountability	Low
Social media platforms	High	Controlling access, proliferation and moderation of digital media	Low

Figure A. Simplified Stakeholder Map

State media broadcasters and independent media companies produce content for the present, hence often not having high interest in archiving their materials by memory institutions. In juxtaposition, how, what, and whether these groups are willing to make their material available holds a considerable amount of power over institutions' archival processes. The inclusion of

social media platforms further complicates these relationships. Although social media will continue to shape how events are documented, access and legality create obstacles as to what can be archived from these platforms. Consequently, while the NIOD may have interest in working with this stakeholder group, they may not share the same interest in archiving what is published.

Affected communities – including refugees, survivors, and diasporic groups – are implicated as record subjects whose digital traces may be collected, contextualized, or withheld. While their control over how they get remembered remains low unless actively involved in the archiving process, their interest is high due to the stakes in contemporary conflict where information's power is escalated. NIOD staff themselves operate as mediators between these groups, translating institutional mandates, ethical considerations, and practical constraints into concrete archival decisions. While NIOD holds substantial power in shaping the archive, this power must be exercised in a balanced manner: staff members must continuously weigh competing stakeholder expectations while operating under legal, technical and resource constraints.

Rather than operating as isolated actors, these stakeholders are bound together through uneven distribution of power, responsibility, and dependency that structure how web archiving decisions can be made. Institutional mandates and funding structures delimit what is possible, while content producers and platform owners exert influence over what can be collected in practice. Researchers and affected communities, although highly invested in the availability and framing of archived material, remains structurally dependent on institutional selection and access regimes. To capture these dynamics, our full stakeholder and tensions mapping (Appendix 1) identifies the key frictions that emerge between completeness, legality, transparency, and ethical care. The simplified model presented here serves as an analytical entry point, while the extended mapping in the appendix makes visible how these tensions accumulate and intersect across institutional, technical and social dimensions of web archiving at NIOD.

Interviews: Institutional perspectives and internal differentiation

During the second and third week of the case study project, our group carried out the semi-structured interviews with two NIOD staff members. Interviews with them reveal that web archiving is not approached as a singular institutional project, but is instead articulated through differing professional perspectives that nonetheless share key assumptions. Both interviews

emphasized that selection is unavoidable in digital environments and that abundance of online material necessitates deliberate choices about scope and relevance (Interview Arvid 2026; Interview Victor-Jan 2026). In this sense, web archiving is not framed as a problem of technical capacity, but as one of institutional judgement under conditions of scale and constraint.

Through our interview with Arvid, we understood that web archiving is situated within a broader set of current practices through which NIOD engages with contemporary conflicts and their societal impact. These include conducting interviews with affected communities, establishing forms of "safe havens" for vulnerable materials, and exploring web archiving as a contemporary third approach to documentation (Interview Arvid 2026). In this framing, web archiving is not a replacement for existing methods, but an additional mode of collecting that raises distinct legal, ethical, and procedural questions. The primary concern here lies not in collecting web-based material per se, but in doing so transparently and responsibly. Selection decisions, search terms, and access conditions therefore need to be explicitly documented, as the institution assumes responsibility once material is preserved and potentially made accessible (Interview Arvid 2026).

By contrast, the Head of Collections emphasizes continuity with existing analogue practices, particularly the curated newspaper clipping collections. As opposed to viewing web archiving as a fundamentally new undertaking, this view frames it as an extension of established topic-based selection logics, adapted to digital environments. The use of "hotspots" – specific events or conflicts deemed societally significant – was highlighted as a pragmatic way to delimit scope while preserving content likely to be valuable for future research. Additionally, active selection is labour-intensive (Interview Victor-Jan 2026). Within this view, web archiving does not necessarily require extensive new frameworks but can build on familiar curatorial judgement focused primarily on content instead of comprehensive contextualization.

These differing emphases do not represent opposing positions but reflect complementary institutional logics. Both interviewees acknowledged sustainability as a key constraint, understood not primarily in terms of storage capacity but as long-term stewardship, including maintenance, migration, and infrastructural dependency (Interview Arvid 2026; Interview Victor-Jan 2026), Similarly, both recognized that legal and privacy considerations impose limits on what can be collected and made accessible, particularly when dealing with material related to ongoing conflict and violence (Interview Arvid 2026).

In combination, the interviews demonstrate that web archiving at NIOD is negotiated across multiple internal rationales, balancing transparency and ethical accountability against

pragmatic selection and continuity with existing practices. These internal differences make visible how archival responsibility is not centralized but distributed, and how decisions emerge through ongoing negotiations rather than predefined policy.

Key Tensions Structuring Web Archiving at NIOD

In this section, we underscore the tensions drawn out from the diagram in Appendix 1, institutional context, and interview insights to indicate that web archiving at NIOD is structured by a set of recurring tensions. These tensions mould how decisions about scope, selection, access, and documentation are negotiated in practice. Hence, they persist across different stages of the archival process and accumulate over time.

The first tension emerges between **abundance and selection**. Digital environments produce an overwhelming volume of web-based material related to contemporary conflicts, creating expectations of comprehensive documentation. Both interviewees emphasized, however, that digital abundance does not eliminate the need for selection but instead intensifies it (Interview Arvid 2026; Interview Victor-Jan 2026). Selection becomes unavoidable not because of technical scarcity, but due to institutional mandates, research relevance, and long-term stewardship imposing limits on what can be responsibly preserved. In this line of thought, selection is not a failure of capacity but an inherent condition of web archiving.

Secondly, another point of tension concerns **visibility and vulnerability.** Web archiving increases the visibility of individuals, communities, and narratives that may otherwise remain undocumented, particularly in relation to conflict, displacement, and trauma. At the same time, increased visibility can expose record subjects to harm, misrepresentation, or unwanted future scrutiny. This tension was articulated most clearly in discussion of privacy and responsibility, where interviewees stressed that NIOD assumes ethical responsibility once material is preserved and potentially made accessible (Interview Arvid 2026). Visibility, therefore, cannot be treated as an unqualified good but must be weighed against the vulnerability of those represented in web-based records.

The tension between **speed and care**, are the third divergence we identified. Contemporary conflicts unfold rapidly online, generating pressure to collect material in near real-time before it disappears. However, ethical reflection, legal assessment, and contextualization require time. Both interviews highlighted that while it may be technically possible to collect quickly, responsible documentation demands careful deliberation, particularly when dealing with sensitive material (Interview Arvid 2026; Interview Victor-Jan

2026). This creates a structural tension between timely intervention and ethical restraint, in which acting too slowly risks loss, while acting too quickly risks harm.

Lastly, a persistent tension exists between **privacy and transparency**. Transparency is widely recognized within NIOD as essential for accountability, research integrity, and public trust. As stated previously, documenting selection criteria, search terms, and methodological choices enables critical engagement with archival collections (Interview Arvid 2026). Concurrently, legal and privacy obstacles limit what can be disclosed about both the content of collections and the individuals represented within them. Transparency, in this sense, does not eliminate ethical risk but operates as a form of accountability that must coexist with necessary restrictions.

These tensions do not function independently. Decisions made to address one tension often intensify another; increasing selection scope may reduce visibility, prioritizing speed may undermine care, and expanding transparency may conflict with privacy protections. The diagram in Appendix 1 visualizes how these tensions intersect and compound across institutional, technical, and ethical dimensions of web archiving at NIOD. Thus, we do not treat the project as resolving these tensions, instead treating them as a constitutive of web archiving practice. They thereby form an analytical basis for the decision simulator, where choices are modelled not as right or wrong solutions, but as trade-offs whose consequences emerge cumulatively over time.

Analysis: Decision-Making in Web Archiving

The preceding section show that conceptualizations of web archiving at NIOD are structured by persistent tensions and negotiated responsibilities. In this section, our analysis examines how those pressures translate into concrete institutional decision-making around scope, collection methods, access, and documentation. Drawing on our case study work – stakeholder/tension mapping, interviews with NIOD staff, and the iterative translation of those insights into a decision simulator – we treat web archiving as a practice of institutional judgement that unfolds through sequences of choices.

Scope Definition and Collection Strategies

At NIOD, decisions about scope form the foundation for potential web archiving initiatives. In our interviews, scope was consistently framed as a matter or responsibility under restraint: institutional mandate, anticipated research value, legal obligations, and long-term stewardship

all delimit what can be collected and maintained (Interview Arvid 2026; Interview Victor-Jan 2026). The recurring reference to "hotspots" is significant here. Hotspots operate as a pragmatic scoping device that links NIOD's mandates to public controversy, societal impact, and Dutch entanglements in transnational conflicts. In our case study discussions, hotspots surfaced as a way to make selection criteria explicit without claiming exhaustiveness, and to align collecting with existing NIOD logics of topic-based curation (Interview Victor-Jan 2026). At the same time, this tool, like other thematic/topic-based collecting approaches, introduces structural limitations. By privileging identifiable events and highly visible platforms, hotspots risk reproducing existing media hierarchies, compressing ethical reflections, and stabilizing present-day institutional framings as future archival structure (Brügger and Schroeder 2017). Thus, institutions need to negotiate how to work with these risks, rather than treating hotspots as a neutral solution.

Collection strategies operationalize scope through specific tools and workflows. Automated crawling, keyword harvesting, and scheduled capture offer scalability and consistency, but they also shift where appraisal happens; decisions get embedded in technical parameters such as seed lists, keyword sets, capture frequency, and crawl depth (Brügger and Schroeder 2017, p. 10). Milligan (2019) similarly notes that scale does not remove appraisal but relocates structural choices that outline what can be later studied (p.16). Manual curation enables contextual sensitivity and case-specific judgement yet increases labour requirements and risks narrowing selection to what is visible to staff at a given moment. In our mapping exercise, this surfaced as a recurring institutional trade-off: collecting at scale tends to compress description and context, while curatorial depth constrains coverage. The interviews reinforced that hybrid approaches are often considered, but they introduce governance questions about alignment: which judgements are delegated to automation, which remain under human responsibilities, and to what extent are inconsistencies documented (Interview Arvid 2026).

Algorithms, Crawlers, and Infrastructural Judgement

Algorithms and crawlers function as more than neutral infrastructures; they actively shape archival outcomes. Milligan (2019) makes this point in a way that directly resonates with web archiving's technical mediation, arguing that "algorithms lie at the heart of almost all of research with digital resources," and calling for historians to "open up the 'black boxes' of proprietary or opaque search engines" (p.21). In the context of web archiving, this has two

implications: first, collection tools determine what is captured and what is missed; second, access interfaces and search systems mould what users can later find and interpret.

Our interviews echoed this concern in institutional terms. Automated capture is attractive because it supports manageability and long-term stewardship – potentially more sustainable and faster than human curation —, yet NIOD staff noted that tools can be "blind" to historically significant materials if selection logic is delegated too fully to technical criteria (Interview Arvid 2026). Crawlers privilege what is linkable, text-based, and technically accessible, while marginalizing content that is platform-restricted, frequently updated between crawls, or embedded in proprietary formats (Milligan 2019). These infrastructural effects are not incidental: they constitute selection by other means. This is why algorithmic and tool-level decisions should be treated as part of archival appraisal and documented as such.

Documentation and Metadata as Archival Practice

As NIOD ventures deeper into web archiving, documentation becomes a primary mechanism to make archival decisions accountable over time. During our interviews, transparency was repeatedly articulated not as a claim to neutrality, but as a practice of recording what was done and why: documenting scope boundaries, selection pools, keyword lists, exclusions, and access conditions (Interview Arvid 2026: Interview Victor-Jan 2026). In our own cause study workflow, this emphasis became concrete when we attempted to translate interview insights into "decision nodes" for the simulator: it quickly became clear that without documenting why a choice was made (and what was not captured), later interpretation becomes misleading, even if the underlying collection is technically sound.

Metadata is central to this form of accountability. Descriptive, technical, and administrative metadata collectively determine whether an archived collection remains intelligible in ten or twenty years. Recording crawl parameters (frequency, seed URLs, depth), time boundaries, versioning limits, and selection rationales shape what future users can infer about absence, bias, and representational scope. This links directly to a literacy problem identified by Schader and Winters (2021): "the question is not only one of being able to access the resources but also of being able to understand them … they require comprehension of both their shaping and their contextualization" (p.141). Accordingly, metadata is not auxiliary; it is part of the interpretive apparatus that makes web archives usable as evidence.

Data Envelopes and Managed Transparency

In our discussions with Arvid, data envelopes were raised as an idea for balancing transparency with ethical restraint (Interview Arvid) 2026; and previous to the interviews, in our group discussions, as a practice for making visible what is not there. Data envelopes provide a way to document the existence, scope, provenance, and access restrictions of collected material without exposing content that could cause harm or violate legal limitations (Luthra and Eskevich 2024). This is especially relevant in web archives of conflict, where record subjects did not anticipate long-term preservation or future reuse of their digital traces.

Data envelopes also sharpen the role of metadata as governance. They enable NIOD to communicate what exists, under what conditions it was collected, what limitations apply – such as copyright, privacy, platform restrictions – and what access pathways exist – such as on-site only, request-based, or embargoed. The value of this approach became apparent in the gap between two internal emphases: one interviewee prioritized transparent procedural guidelines, while the other treated digital collecting as analogous to the newspaper clippings model focused on preserving content for future research (Interview Arvid 2026; Interview Victor-Jan 2026). Data envelopes help reconcile these positions by allowing NIOD to preserve content while still documenting context and constraints in a visible, standardised way.

Although data envelopes provide an important mechanism for documenting selection criteria, technical restrictions, and institutional rationales, they also have clear limitations. Transparency through documentation does not eliminate ethical risk or structural bias, nor does it resolve tensions around exclusion, privacy, or power. Instead, data envelopes function as a form of accountable disclosure, enabling critical interpretation without guaranteeing ethical outcomes. Their effectiveness further depends on users' capacity to access and interpret contextual information, and they may struggle to scale within high-volume web archiving environments. As such, data envelopes should be understood as a necessary but insufficient component of responsible web archiving practice.

Access Regimes, Collaboration, and "Sharing Zones"

Access decisions are inseparable from responsibility. Both interviews stressed that NIOD's obligations intensify when material is made accessible, especially where personal data or sensitive content is involved (Interview Arvid 2026). Access regimes therefore become part of archival design: open access, gated request systems, on-site-only consultation, or delayed release each distribute risk and responsibility differently.

Access also connects to collaboration and interoperability. Schafer and Winters (2021) frames this a developing frontier: "the concept of 'sharing zones'... applies in the web archiving world, notably with regard to gateways and interoperability: transnational metadata and derivative data repositories are probably the new frontier … to ensure that web archives meet researchers' needs" (p.142). This resonates with NIOD's institutional embedding in wider Dutch digital heritage infrastructures, and with interview remarks about the need to clarify NIOD's distinct function relative to other institutions. In our case study framing, "sharing zones" can be treated as a governance option that supports collaboration through metadata exchange or derivative datasets while allowing NIOD to maintain curatorial framing and access restrictions appropriate to subjects such as mass violence and trauma.

Sustainability, Infrastructure, and long-term stewardship

A central structuring condition in NIOD's approach to web archiving is sustainability, both increasing and limiting what can be collected, how it can be maintained, and under which conditions it remains accessible over time. In contrast to popular assumptions that digital storage is inexpensive and therefore unlimited, both the literature and the interviews emphasize that sustainability concerns long-term stewardship rather than short-term capacity. As Terry Cook (1997) notes, in digital environments archival challenges shift away from material decay toward "migration, preservation, and sustained decision-making over time" (p.45). Sustainability is thus inseparable from ongoing institutional responsibility.

This understanding was echoed consistently in the interviews. Both interviewees stressed that while storage itself may not be prohibitively expensive, long-term maintenance, format migration, and infrastructural dependency are significant constraints (Interview Arvid 2026; Interview Victor-Jan 2026). NIOD's reliance on external infrastructures, such as the KNAW e-depot, enables digital preservation but also embeds web archiving within broader organizational and financial frameworks beyond NIOD's direct control. Sustainability therefore functions not as a purely technical issue but as an institutional one, linking archival ambition to available resources, expertise, and governance arrangements.

Sustainability also shapes selection practices directly. As the Head of Collections noted, attempting to collect "everything" is neither necessary nor desirable; instead, selection prior to collection is viewed as a pragmatic way to reduce long-term burdens (Interview Victor-Jan 2026). This aligns with Milligan's (2019) observation that digital abundance intensifies the need for selectivity, as "finding something specific among the available information is the real

challenge" (p.9). Hence, sustainability reinforces selection as a responsible archival practice rather than a limitation.

Simultaneously, sustainability interacts with ethical considerations. Acting quickly to capture volatile web material may secure content, but without sufficient resources for documentation, contextualization, and access management, such intervention risks becoming unsustainable in the long term. As Milligan cautions, web archives are "necessarily incomplete" and shaped by practical and ethical limits (p.16). Sustainability, therefore, does not merely limit scope; it shapes the tempo, scale, and durability of archival decisions. Thus, at NIOD it functions as a cumulative condition that mediates between ambition and responsibility; underscoring that web archiving is not a one-time intervention but an ongoing institutional commitment, where decisions made in the present continue to generate obligations well into the future.

Making the Archive: A Web Archiving Decision Simulator

Rationale

One of the core issues our project group encountered was how to get to a point where we can arrive at a coherent standpoint on web archiving that could be presented to the NIOD as a definitive, recommended strategy for how their collection should be shaped. This complexity stems not from a lack of guidance from staff at the NIOD involved with their web archiving ambitions, nor from a lack of interaction with the academic literature on (web) archiving. It instead flows forth from a recurrent struggle to identify a clear-cut 'correct' strategy to web archiving.

Based on the literature review and on-site interviews with staff-members, our insights instead indicate that there exist a range of tensions in viewpoints between the theory and what is feasible or preferable in practice. Additionally, tensions also exist within the collections department of the NIOD itself. This is not to say that one entity is to be considered right or wrong, but rather to highlight that there is no consensus on what is to be considered as a correct response to the need for web archiving. This resonates with Terry Cook's explanation of archival decision-making as contingent on internal and external influences that shape how archivists act. These influences span a broad network of personal, educational, institutional and socio-political values that are (implicitly) integrated into any type of decision made in an archival context (Cook, 2013, 101-102).

Within the context of web archiving, this lack of consensus is further perpetuated. For instance, Niels Brügger, conceptualises web archives as constructed. This construction process involves a myriad of decisions to be made within a complex environment regarding selection, collection, preservation and accessibility. These decisions ultimately lead to a representation of the online web which differs from how we normally encounter it, which he frames as a "reborn" web (Brügger, 2018, 73-74). While these decisions can be separated for analytical purposes, in practice they compound and interact, leading to a collection that is influenced by technological, institutional and other contextual factors (Ibid).

Based on these insights, it becomes clear why it is difficult to arrive at a definitive web archiving strategy in practice. Because archival decision-making is shaped by personal, professional, institutional and societal factors and these choices are compounded and interrelated, the challenge is not necessarily identifying a 'correct' way of archiving the web but rather understanding different strategies and their outcomes.

Meanwhile, there remain so many possibilities that it becomes increasingly difficult to understand how each scenario plays out through text alone. We therefore set out to visualize how archival decision-making can produce various scenarios when designing a web archive. Our response to this challenge was to develop a decision tree model which navigates different recurrent concepts identified within the literature and interviews.

Design

Our decision tree model was developed as an interactive simulation tool, inspired by narrative-based videogames, where discrete decisions lead to unique end scenarios. The simulator formally titled, *Making the Archive: A Web Archiving Decision Simulator*, was developed in Twine, an open-source software tool that allows users to build interactive, non-linear stories (Twine, n.d.). While Twine allows for a wide range of technical possibilities through programming, it remains relatively accessible to users with limited coding experience. By relying on hyperlinking structures between different passages that correspond to discrete decision-points, Twine enables the production of branching narrative structures that resemble a decision tree.

The simulator was designed to prioritize conceptual clarity and over complexity. The narratives are deliberately kept as concise as possible, ensuring that attention is focused on the decision-making process and promoting reflection on the decisions made. This approach reflects the simulator's primary purpose; it is intended to be played by interested parties at the

NIOD and invite them to discuss and reflect on their intended web archiving practices as well as introducing alternative ways of approaching this challenge.[1]

We do not claim that this simulator provides an all-encompassing overview of possible web archiving workflows. As such, to support transparency and further re-use of the simulator, we have decided to host the HTML-file openly on GitHub (GitHub, 2026). There it can be freely accessed, checked and downloaded. In doing so, NIOD and other interested parties have the ability to further expand the simulation beyond its current iteration.

Value System

Our first objective was to maintain a scoring system that tracks different budgets related to financial capacity, technological infrastructure and the amount of human labour required in the form of tokens. This was intended to showcase how different approaches would constrain the organisation in material ways. However, we came to the realization that this would lead to a relatively one-dimensional experience, because it could lead to situations where players would end up shifting their choices to meet the budgetary limits.

We therefore chose to update the scoring system to track archival and institutional values instead. The scoring system operationalises five different value dimensions, namely: ethics, sustainability, legality, transparency and completeness.

- Ethics in this simulation refers to the extent to which the institution takes on responsibility towards individuals and communities that are represented in the archive.
- Sustainability refers to the extent to which the institution is able to maintain, preserve and manage the archive over a long-term period.
- Legality refers to the extent to which the archive complies with relevant legal frameworks.
- Transparency refers to the extent to which archival actions are documented and communicated to its audience.
- Completeness refers to the breadth and coverage of the collection, including the extent to which diverse perspectives are included, while simultaneously acknowledging that a fully complete collection is not possible.

[1] We intend on speaking with a group of interested employees at the NIOD on the tenth of February to report back on our findings. Our intention is to introduce the simulator to them and allow them to play through it. Based on their outcomes, we will then encourage the group to discuss their findings and what trade-offs are most pressing to them and how different strategies might align with institutional mandates.

Decisions Nodes

Our second design objective was to formulate the decision points that end-users would engage with. In the initial design phase, our first decision node consisted of three discrete choices that relate to the selection scope that the web archive would follow. This topic emerged as a recurring discussion point amongst our project group and is similarly reflected in our interviews, where tensions were articulated between what an institution might theoretically wish to capture and what is practically and legally feasible.

Eventually, due to time constraints, we arrived at the conclusion that starting off with three discrete choices leads to a snowballing effect in terms of how many passages would have to be developed. We therefore decided to pursue a story structure where players are faced with a dilemma containing two choices per passage. We also decided upon a four-tiered list of questions that relate to what we consider to be the most pressing issues that web archivists are faced with.

The first question relates to the scope that the institution will set in terms of what materials are to be included in their collection. The dilemma in this instance is between setting a manageable, legally and institutionally aligned scope of collecting digital news articles or choosing to broaden scope and include social media content as well. The second option might lead to more legal inquiries and is less manageable but does allow more variety and representational breadth to be introduced into the archive.

The second dilemma relates to the collection method, where players are faced with how they access sources they wish to archive and how to capture them. Inspired by the literature, which often favours technical approaches, and contrastingly, the way preliminary web archiving projects at NIOD have been performed primarily through human selection and collection methods, we decided to let this inform the choices one has to make. We also elected to add a transparency angle into this question by making the choice for a web crawler lead to struggles with black-box algorithms and having the human-led approach require documentation work through filling out data envelopes.

The third decision node concerns how players wish to expand upon the archive. We conceptualized two approaches that are methodologically different, yet both lead to an increase in completeness. The first decision is inspired by inter-institutional linking initiatives, like that of *Netwerk Digitaal Erfgoed*, which privilege expansion through interoperable platforms and interlinking digital heritage collections through linked open data (Ministerie van Onderwijs, Cultuur en Wetenschap, 2024, pp. 21-22). On the other hand, we came up with the choice to

expand upon the archive through community participation, which reflects the preliminary projects both interviewees have spoken about, and which is a central point in archival literature, as demonstrated above in the analysis.

Finally, there is the dilemma about accessibility, which plays out relatively straightforward, allowing you to either constrain access through a request system or making the archive freely available. These dilemmas and their associated value-changes led to the construction of the following score system:

Choice		A	B		
Scope	1	News Media	News + Social		
Method	2	Crawler + Black-Box	Human curation + data envelope		
Institutional linking - Community participation	3	Interlinking	Community participation		
Accessibility	4	Public	Restricted		
Score system	Ethics	Sustainability	Legality	Transparency	Completeness
Start	4	4	4	4	4
1A	0	2	2	0	-2
1B	0	-2	-2	0	2
2A	-1	1	0	-1	1
2B	1	-1	0	1	-1
3A	-1	1	0	0	1
3B	1	-1	0	1	1
4A	1	-1	-1	1	0
4B	-1	1	1	-1	0

Figure B. Simulator score system.

As can be gleaned from the figure above, the first decision is foundational in terms of setting the stage for sustainability, legality and completeness. This design choice reflects scope as a foundational orientation in web archiving, what type of material is allowed entry has immediate and significant implications for how manageable and legally-aligned the archive will end up being. Therefore, this decision is weighted as the point through which the most value is gained or lost across the aforementioned dimensions, thereby establishing it as the basis within which later decisions operate.

The scoring system further illustrates that later decisions operate less as foundational shifts, but rather as value adjustments that refine the collection's orientation. For example, in the third decision node completeness increases regardless of which option is selected. This reflects our conceptualization of completeness as representing both volume and representational variety, as expansion can be brought about through interlinked institutional collections as well as through community participation efforts. Meanwhile, these choices still have discrete implications for ethics, sustainability and transparency. In doing so, the scoring

system makes explicit that the simulator does not treat values as universally compatible but simulates how increasing one value dimension will still require trade-offs in other dimensions.

End Scenarios

The scoring system is crystallised into sixteen different end scenarios that leave the player with an overview of their scores for each value, alongside a title and short description for their approach. Next to these descriptive and numeric evaluations, we also wanted to highlight what trade-offs had to be made to come to the collection that each player has designed. We therefore included sixteen different spider charts containing each of the five values, using the end-scores to model them. We elected not to include a rangefinder so that the models would remain more open to interpretation, rather than functioning as purely a reflection of numeric values.

For most of the graphs the trade-offs are easily visible. Yet, as visualized below, end scenario VIII, which follows the pathway of AABA, the values are relatively close to one another. This can give the illusion that the archive excels at representing each of the values rather well. In practice, however, the values are all relatively constrained compared to other scenarios, meaning that the strategy reflects balance through limitation, rather than through optimisation. Therefore, the simulator tries to demonstrate that balance does not necessarily indicate an absence of trade-offs, but may also reflect the effects of cumulative constrained decision-making.

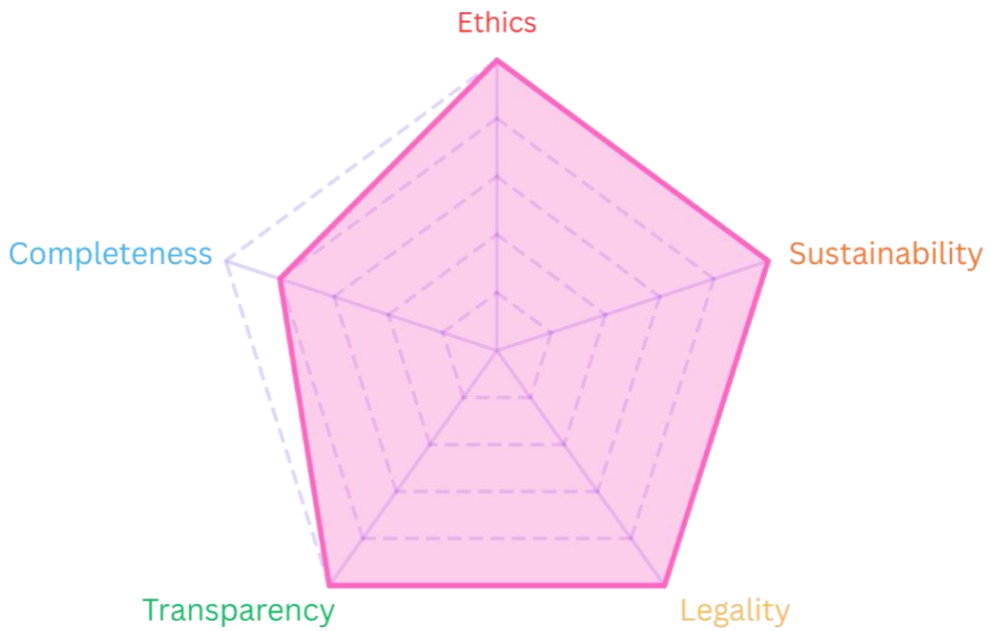

Figure C. Spider graph for end scenario VIII.

Limitations

As mentioned above, the simulator was developed within a limited project timeframe, which required several trade-offs regarding its design. First and foremost, in this regard is that our initial design was far broader in scope. We envisioned a model where there would have been a lot more choices available per decision node. However, due to the exponential character of a branching narrative structure, this was too complex to implement given the fact that we wished to contextualize each decision and end scenario to some extent. This means that we have constrained the number of dilemmas. We also integrated two dilemmas into decision node two, combining the collection method with transparency quandaries.

Another limitation is the fact that the decision tree is currently relatively static, there is no full interdependence of certain questions. For instance, we conceptualized making the accessibility question dependent on the scope choices made, whereby a broader scope would lead to the player being unable to make their collection freely accessible. If the simulator were to be developed further, this is a mechanic that would be interesting to explore further. On the other hand, for our purposes of eliciting reflection and discussion around different approaches, we believe that our current design philosophy is more beneficial, as it enables players to make decisions that might not be practically feasible, but do respond more to the priorities of the player and thereby stimulate discussion.

A further limitation is that the value system we came up with is heavily dependent on our interpretations of (web) archiving literature and impressions of how the NIOD functions. While this is inevitable and aligns with contemporary understandings of archiving in general, there are multiple ways to argue value changes for each decision based upon one's positionality. Although this is a crucial component of our design philosophy, the value system could be expanded upon by other archivists and external stakeholders, so that it would reflect a broader set of viewpoints on web archiving matters. This is why we have included a set of reflexive questions at each end scenario, explicitly asking for feedback on the value system and whether there are other possibilities that were left unexplored. We hope that by making the files freely available, others will be inspired to further develop the tool.

Finally, we entertained the idea of implementing some type of analytical feature that would keep track of the most travelled pathways. We are therefore unable to get insights into what would be the most desirable options for the institution. While this would be an interesting addition to add during future development, the simulator's primary aim remains reflective, rather than evaluative.

Intended Use

In conclusion, the simulator was designed as a reflective tool that supports institutional discussions around their web archiving strategy. It is intended to be used collectively, for instance in internal meetings or in workshop settings. Staff members are able to play through the simulation and during or afterwards, compare their responses to decisions, trade-offs and outcomes. By externalising decision-making into an interactive format, we hope to provide a shared language through which different perspectives and priorities can be articulated more clearly. In doing so, the tool may help reduce uncertainty in strategic discussions by translating abstract tensions into a structured set of choices and outcomes.

Importantly, this simulator is not intended to replace policy documentation or professional expertise, nor does it claim to produce definitive recommendations for a web archiving infrastructure. Its value lies in facilitating conversations around perceived trade-offs between different approaches and making them visible. Due to its open access approach, it also stands to be expanded upon in future iterations, depending on institutional priorities.

Outcomes and Implications

Within the report, our analysis demonstrates that web archiving at NIOD cannot be resolved through a stable policy framework. Rather, it produces a set of institutional implications that concern how responsibility is distributed over time, how decisions accumulate, and how archival practice is made accountable under conditions of uncertainty. The outcomes of our project therefore do not take the form of definitive solutions, but of clarified consequences and procedural insights that can guide responsible institutional action. Here we identify five outcomes and implications from our findings, which ultimately shapes our advisory section to NIOD.

Firstly, archival responsibility is not gained or broken from a singular decision, but built up by sequences of choices made throughout the archival process. Decisions regarding scope, collection methods, access, and documentation do not operate independently. Therefore, choices made early in the process – such as defining hotspot, selecting capture tools, or delegating appraisal to automated systems – shape the range of options available later. This sequential logic underscores that no single decision is catastrophic in itself; ethical risk, sustainability pressure, and representational bias arise through the accumulation of decisions across the archival lifecycle. This insight directly informed the decision nodes of the decision simulator, where consequences become visible only through extended decision paths.

Completeness, ethical care, sustainability, legality, and transparency operate in tensions as opposed to harmony, as a result, their maximization cannot be done simultaneously. Reflecting on our simulator, attempts to expand coverage tends to compress documentation and contextualization; efforts to increase transparency may conflict with legality obligations; prioritizing sustainability through technology may result in rapid capture but can also undermine ethical deliberation. The constant interplay between the values, positions the player to reflect on how pulling one string, also pulls on another. These tensions should not be interpreted as institutional failure, but as structural limits inherent to web archiving under conditions of digital abundance and ephemerality. Recognizing these limits enables NIOD to frame selectivity as responsible judgment rather than deficiency.

Another implication refers to how documentation turns into a critical mechanism of accountability, not a guarantee of neutrality. Practices such as metadata enrichments, procedural documentation, and the use of the data-envelopes do not eliminate bias, harm, or exclusion. However, they enable institutions to explain how and why decisions were made, and to render absences and constraints legible for future users. In our interviews with the NIOD staff, we learned that clear documentation of past guidelines is often lacking in analogue collection, complicating later interpretation and accountability (Interview Arvid 2026; Interview Victor-Jan 2026). As the use of technical systems and algorithms increases the 'black box' effect (Milligan 2016), transparency in this sense functions as a form of institutional self-positioning: it allows NIOD to take responsibility for interventions without claiming objectivity or exhaustiveness. Ultimately, in our view this means that the NIOD needs to take documentation into consideration before beginning web archiving, if they want to create an archival process that allows for a deeper involvement of other stakeholder groups.

With regards to World War II archival material, the NIOD actively receives material and as they possess an extensive collection, they often practice selectivity. As a result, selectivity and topic-based selection may not be a new venture, but in terms of contemporary conflicts we learned from one of our interviews that NIOD needs to now seek out material without being flooded with donations. Hence, our analysis indicates that web archiving shifts NIOD's role from a primarily retrospective archive, towards an active participant in contemporary knowledge production. This requires the NIOD to form connections and start reaching out to other stakeholders. By intervening in live digital environments, NIOD moves closer to shaping how present-day conflicts can be studied, remembered, and interpreted in the future. This transition heightens ethical and governance responsibilities, particularly in relation

to affected communities whose digital traces may be preserved without anticipation of archival reuse.

Finally, we highlight that ethical intent is constrained by infrastructure and resources. Even well-motivated archival decisions are moulded by legal frameworks, technical systems, staffing capacity, and external dependencies. Recognizing these obstacles does not resolve responsibility but situates ethical care within the realities of institutional practice. Drawing on Caswell and Cifor's (2016) notion of radical empathy, we suggest that ethical care in web archiving does not reside in benevolent intent or comprehensive preservation, but in the recognition of relational obligations toward record subjects, communities, and future users. At NIOD, this responsibility is materially restricted by legal frameworks, infrastructural dependencies, and resource limitations. While the institution archives with researchers in mind, their involvement or access to the process of archiving seems to remain unclear. As a result, radical empathy comes up not as an idealized ethical stance, but as a situated practice enacted through restraint, documentation, and selective absence. In this perspective, ethical care is exercised through conscious decisions about what should not be made visible or accessible.

Ultimately, these outcomes support our advisory position that suggests process over prescription. Instead of proposing a fixed web archiving strategy, our project offers NIOD a framework for initiating web archiving as a reflective, documented, and sustainable institutional practice – one that acknowledges trade-offs, makes decision-making visible, and treats responsibility as an ongoing commitment rather than a resolved problem.

Procedural Starting Points for Web Archiving at NIOD

Based on all our analyses, we do not recommend that NIOD should adopt a single, fixed web archiving strategy or aim for comprehensive capture. Instead, we suggest approaching web arching as a gradual, reflective institutional practice that develops through documented decisions, experimentation, and ongoing evaluation. The following procedural starting points outline how NIOD could begin engaging with web archiving in an accountable and manageable way.

1. <u>Focus on clearly bounded scope, not platform-specificity</u>

NIOD could begin framing web archiving with a limited number of topic- or event-based hotspots. These should be explicitly defined in relation to NIOD's research mission, societal relevance, and available capacity. Previously, NIOD has ventured into some practices of this

such as with the Maccabi riots and Central Archives of Special Jurisdiction (CABR) cases (Interview Victor-Jan 2026). Based on our findings, we support this approach although we encourage the NIOD to broaden who gets to decide on what is defined as a "hotspot." The decision process should include diasporic groups, activists, and researchers to provide multiple contextualizations of events. This could start with inviting participants to take part in interviews which the NIOD conducts and preserves. Importantly, scope definitions should include not only what will be collected, but also what is excluded and why. By making these boundaries explicit, it helps prevent later assumptions of completeness.

2. Documentation should be a core practice from the outset

Decisions about scope, selection, tools, and access should be documented as part of the archival process itself, not as an afterthought. Practices such as structured metadata, procedural notes, and data-envelopes can help make these decisions legible over time. Further, inviting evaluations from outside departments and the wider institution would invite a reflexive, evolving process that abandons ideas about fixed methods. Thus, documentation does not release from bias or ethical risk, but rather gives NIOD the opportunity to explain and invite discussion over how collections are structured and under which constraints they were created.

3. Dedicate a group of NIOD staff for web archiving

Taking on web archiving contemporary events – even material related to WWII – requires a collaborative effort. Milligan (2016) recognizes the role of researchers in how history is being interpreted and stresses the need to acknowledge their part in the archival process. Caswell and Cifor (2016) call for archivists to be caregivers and we argue that this is fostered through community and collaboration. We understand that it may be difficult to initially set up such an environment and therefore we prompt NIOD to begin with forming a group of NIOD staff and researchers. This group can start to test different combinations of manual selection and automated tools, as well as to assess legal, ethical, and sustainability implications in practice. Such collaborative approaches create space for learning without locking the institution into long-term commitments too early, as well as enabling evaluations from outside perspectives.

4. Engage in building internal alignment through shared deliberation

Our interviews indicate that perspectives on web archiving differ within NIOD, particularly regarding transparency, contextualization, and continuity with existing practices. In addition to policy alone, to resolve these differences NIOD could use internal workshops or scenario-based

exercises to surface assumptions, trade-offs, and responsibilities. Along with fostering collaboration, external workshops would also provide a site to deliberate and share perspectives, learning from institutions who have ventured into web archiving initiatives. In terms of scenario-based exercises, we hope our simulator showed the benefits of conceptualizing before doing. By collectively experiencing trade-offs in institutional values, it offers an opportunity to reflect and discuss how sequences of decisions result in different outcomes. This supports shared understanding without requiring immediate consensus.

5. Combine access decisions with ethical responsibility

Access should be treated as a design choice rather than a default outcome of collection. NIOD may consider differentiated access regimes – such as embargoes, on-site consultation, or request-based access – especially for sensitive web material related to ongoing conflict and violence. From a feminist ethics of care perspective, access decisions are not merely legal or technical matters, but relational practices that shape how responsibility toward record subjects and affected communities is enacted over time (Caswell and Cifor 2016). This suggests that access regimes should not be fixed solely through internal institutional judgement, but periodically revisited in dialogue with relevant stakeholders, including researchers, community representatives, or activists with contextual expertise. Such engagement does not transfer responsibility away from NIOD, but broadens the knowledge base through which care, risk, and potential harm are assessed. In this way, access becomes a revisable practice of care rather than a one-time decision, allowing NIOD to remain responsive as social, political, and ethical conditions evolve.

These starting points do not resolve the tensions outlined in this study. They deliberately leave certain questions open, including how long-term governance structures should evolve, how affected communities can be meaningfully involved over time, and how institutional documentation practices will be interpreted by future users. These limitations are not shortcomings of the approach, but reflect the structural conditions under which web archiving operates. Overall, the recommendations suggested by us emphasize that responsible web archiving at NIOD is less about finding the "right" system but more about establishing conditions under which decisions can be made transparently, ethically, and sustainably. Web archiving, hence, becomes an ongoing institutional commitment.

Conclusion

Initiating web archiving practices presents possibilities, challenges, and institutional implications for the NIOD Institute for War, Holocaust and Genocide Studies. This research has shown that when web archiving is understood as an institutional process shaped by selection, responsibility, and governance, it cannot be resolved through a stable policy framework alone. The outcomes of our project have therefore not taken the form of definitive solutions, but of clarified consequences and procedural insights that can guide responsible institutional action.

The case study project assignment was deliberately broad, this meant that throughout the research process we had to set our own outlines and boundaries. Through stakeholder analysis, interviews and an extensive literature review, it became clear that we could not resolve this into a single, fixed web archiving strategy. Our findings pointed to the structural, ethical, and infrastructural uncertainties that make such closure neither realistic nor necessarily desirable. While this outcome may diverge from initial expectations, it reflects the realities of web archiving as an evolving institutional practice.

Consequently, our analysis and recommendations remain relatively theoretical in nature. But we hope that with our simulator game, research outcomes and procedural starting points, that deliberately leave some questions open, our time and effort will be of use in the process of moving towards web archiving. We would like to thank everyone at NIOD for their time and help in supporting this case study, providing guidance and placing their trust in us.

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Appendix 1

Stakeholder mapping

Our first stakeholder map, as shown down below, is based on our first meeting at NIOD as well as online research. In this model we show, different than what was required, not just the stakeholders within NIOD, but a broader selection of societal and organisational stakeholders. Additionally, we visualised relations and tensions that we could identify between them, with the arrows representing power dynamics. Based on this mapping we created a different version that includes power, roles and interests of the different stakeholders in a more simplified template. This version is used and explained in the case study paragraph of the report.

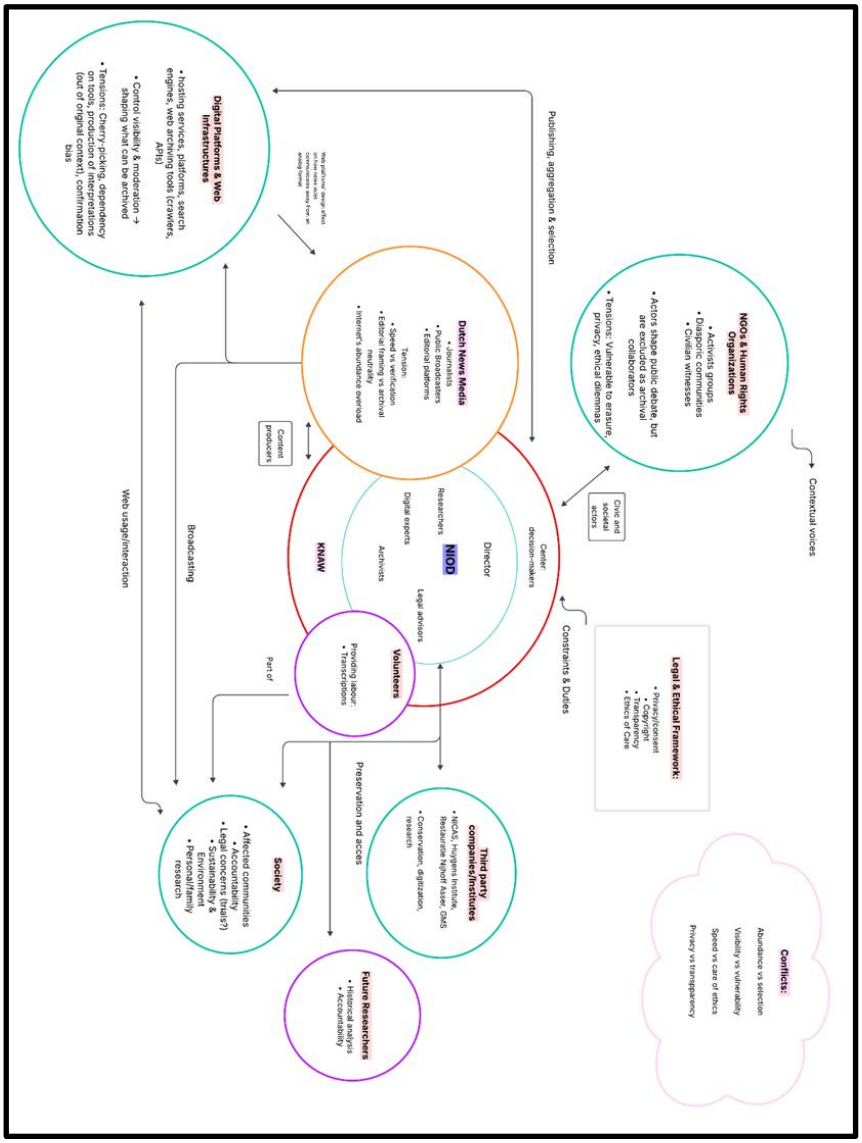

Figure D. Stakeholder-Mapping of NIOD including relations and tensions.

Appendix 2

Interviews

Arvid de Raaij is a staff member of NIOD's collections and services specialising in digital collections. His activities include digitalisation and working with born-digital materials. Arvid was our supervisor during this project, but we interviewed him in his role as digital collections specialist. This conversation provided insights into current collection strategies at NIOD and how these reflect the division of the collection and research departments within the organisation. Because Arvid is one of the people working on expanding the collection to include current day conflicts it was valuable to get his insight into the possibilities and problems that arise from web archiving at NIOD.

Victor-Jan Vos is the head of the Collections and Services Department at NIOD. His responsibilities include the management, storage, and access to NIOD's archives. He focuses on improving digital access and sustainable storage of the collections. Our conversation with Victor-Jan provided us with views on how and within what context web archiving could be possible at the organisation. It also showed how this relatively new endeavour of web archiving could fit into a long tradition of archival practices at NIOD. As one of the leading figures within the organisation and especially its collection, it was valuable to gain insight from Victor-Jan into topics such as selection, transparency and collaboration.

Appendix 3

Future Scenario(s)

Rather than presenting a single 'best-case' future scenario, we have approached the future scenario exercise primarily through the simulation tool we developed. The sixteen end scenarios produced by the decision path represent distinct strategic orientations that the NIOD could plausibly adopt, each shaped by distinct priorities with regards to ethics, sustainability, legality, transparency and completeness.

In this sense, the simulator can function as a structured scenario exercise. This approach allows NIOD to explore multiple possibilities, compare their implications and reflect on which trade-offs are considered acceptable or pressing under different institutional priorities. This aligns with our broader findings that web archiving at the NIOD is best approached as an iterative practice grounded in deliberation and documented decision-making rather than as a problem that can be resolved through a single, fixed strategy.

End Scenarios	Ethics	Sustainability	Legality	Transparency	Completeness
V (AAAA)	3	7	5	4	4
VI (AAAB)	1	9	7	2	4
VIII (AABA)	5	5	5	5	4
IX (AABB)	3	7	7	3	4
XII (ABAA)	5	5	5	6	2
XIII (ABAB)	3	7	7	4	2
XV (ABBA)	7	3	5	7	2
XVI (ABBB)	5	5	7	5	2
XX (BAAB)	1	5	3	2	8
XXI (BAAA)	3	3	1	4	8
XXIII (BABA)	5	1	1	5	8
XXIV (BABB)	3	3	3	3	8
XXVII (BBAA)	5	1	1	6	6
XXVIII (BBAB)	3	3	3	4	6
XXX (BBBA)	7	-1	1	7	6
XXXI (BBBB)	5	1	3	5	6

Figure E. Overview of simulation's end scenarios.

At the same time, the simulator outcomes have also informed a set of procedural starting points for how NIOD could begin engaging with web archiving in a manageable, yet accountable manner. Building on these starting points, we have developed a future scenario that does not describe specific external events or predictions, but outlines a plausible future institutional trajectory. Visualized below in figure F, we have tried to express what a web archiving workflow would look like at the NIOD. The figure explores our suggestions, yet

stays away from making definitive judgements about what individual choices should be made. It instead foregrounds shared deliberation and reflexivity with regards to methods chosen, thereby allowing perceived missteps to be corrected.

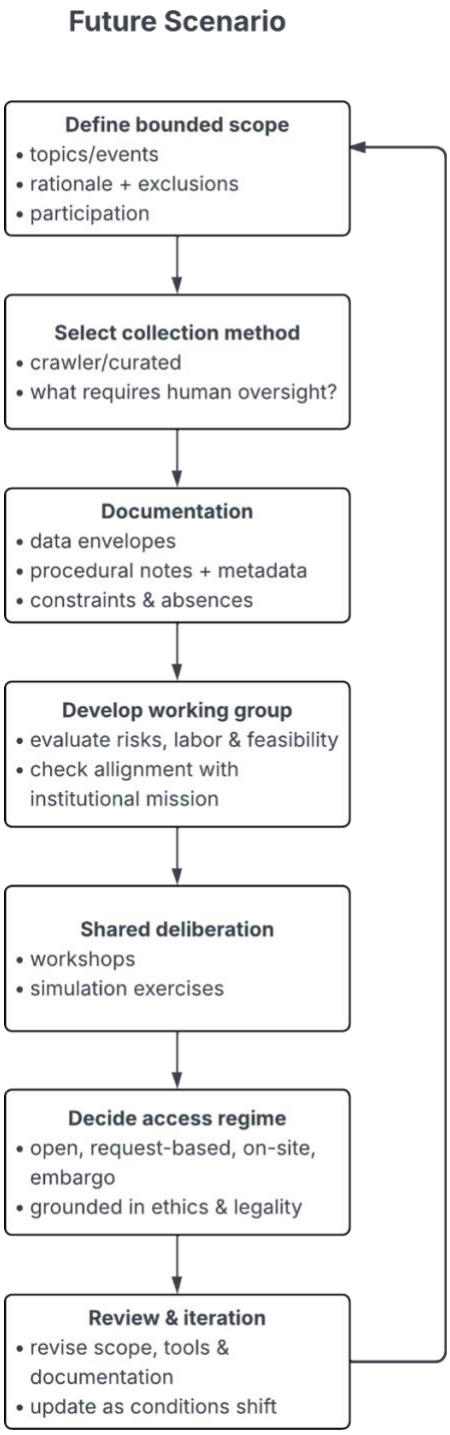

Figure F. Future Scenario.